



THE UNIVERSITY
of EDINBURGH

School of Mathematics

Bayesian Theory Notes

Curtis

Contents

1	Basics of Bayesian inference	4
1.1	Probability Background	4
1.1.1	Events and Probabilities	4
1.1.2	Interpretations of Probability	5
1.1.3	Key Distinction Between Probability and Statistics	5
1.2	Univariate Discrete Distributions	6
1.2.1	Random Variables and Support	6
1.2.2	Probability Mass Function(PMF)	6
1.2.3	Cumulative Distribution Function	6
1.2.4	Mean and Variance	7
1.2.5	Samples, Statistics and Sampling Distributions	7
1.2.6	Parametric Statistical Models	7
1.2.7	Common Discrete Distributions	8
1.2.7.1	Bernoulli Distribution	8
1.2.7.2	Binomial Distribution	8
1.2.7.3	Poisson Distribution	9
1.2.7.4	Key Takeaways	10
1.2.8	Continuous Distributions	10
1.2.8.1	Cumulative Distribution Function	11
1.2.8.2	Probability Density Function(PDF)	11
1.2.8.3	Validity Conditions for a PDF	11
1.2.8.4	Mean and Variance of a Continuous Random Variable	12
1.2.8.5	Quantiles	12
1.2.9	Common Continuous Distributions	12
1.2.9.1	Gaussian Distribution	12
1.2.9.2	Gamma Distribution	13
1.2.9.3	Beta Distribution	13
1.2.9.4	Key Takeaways	14
1.3	Multivariate Distribution	15
1.3.1	Joint Probability Mass Function	15
1.3.1.1	Joint Probability Density Function	16
1.3.1.2	Validity Conditions for a Joint PDF	16
1.3.1.3	Key Takeaways	16
1.4	Marginal and Conditional Distributions	17
1.4.1	Marginal Distributions	17
1.4.2	Marginal Mean and Variance	17
1.4.3	Covariance and Correlation	18
1.4.4	Mean Vector and Covariance Matrix	18
1.4.5	Conditional Distributions	18
1.4.6	Key Takeaways	19
1.5	Bayes' Rule	19
1.5.1	Discrete Example of Bayes' Rule	20
1.5.2	Continuous Example of Bayes' Rule	21

1.5.3	The Role of the Marginal Likelihood	22
1.5.4	Key Takeaways	22
1.6	Introduction to Bayesian inference	23

About these Notes

These notes are based on the course **Bayes Theory**, which follows the framework of *Bayesian Statistical Methods* by **Brian J. Reich & Sujit K. Ghosh**. This course is designed as an advanced undergraduate or foundational postgraduate-level module.

I attended this module at **The University of Edinburgh** during the **2022/23 academic year**. Some time has passed since completing this course, yet I felt compelled to consolidate my original notes into this comprehensive reference to support the wider academic community in their learning and research. If you identify any errors or have suggestions for improvement while reviewing these notes, please feel free to contact me via email at: chenyz9811@gmail.com.

Chapter 1

Basics of Bayesian inference

1.1 Probability Background

Probability theory provides the mathematical foundation for statistical analysis. In probability, we usually assume that a mathematical model for a random process is already known, and we use that model to compute the probabilities of events. For example, if a fair coin is tossed five times, probability theory can be used to calculate the probability of observing five heads.

Statistics works in the opposite direction. In statistics, we observe data and use those data to learn about the underlying process that generated them. For example, if we observe five heads in a row, we may ask whether the coin is fair or whether there is evidence that it is biased.

Therefore, probability and statistics are closely related but have different purposes:

Probability: model $\rightarrow$ data behaviour

Statistics: data $\rightarrow$ model inference

In other words, probability asks what data we should expect under a given model, whereas statistics uses observed data to make statements about the model.

1.1.1 Events and Probabilities

An **event** is a set of possible outcomes from a random experiment. Events are often denoted by capital letters such as

A

The probability of an event A is written as

$Prob(A)$

For example, if A is the event that a coin lands heads, then $Prob(A)$ is the probability of observing heads.

A random variable is usually denoted by a capital letter, such as

X

A particular realised value of that random variable is usually denoted by a lower-case letter, such as

x

For example,

$Prob(X = x)$

means the probability that the random variable X takes the specific value x .

If X is a continuous or interval-valued random variable, we may also write probabilities such as

$$Prob(X \in [a, b])$$

which means the probability that X lies in the interval $[a, b]$.

1.1.2 Interpretations of Probability

There are two common interpretations of probability: the frequentist interpretation and the subjective interpretation.

Under the **frequentist interpretation**, the probability of an event is understood as its long-run relative frequency. If the same experiment were repeated many times, the proportion of times that event A occurred would eventually approach $Pr(A)$.

For example, if a fair coin is tossed many times, the proportion of heads should approach

$$0.5$$

Under the subjective interpretation, probability represents a degree of belief about whether an event will occur. This interpretation is closely associated with Bayesian statistics. For example, an individual may assign a probability to the event that it will rain tomorrow based on available information, experience and judgement.

These two interpretations lead naturally to two major approaches in statistics:

Frequentist statistics

and

Bayesian statistics

In Bayesian analysis, probability is often used to represent uncertainty about unknown quantities, including model parameters. This is why probability theory is essential for understanding Bayesian statistical methods.

1.1.3 Key Distinction Between Probability and Statistics

A useful way to remember the difference is:

Field	Starting Point	Main Question
Probability	A known model	What data or outcomes should we expect?
Statistics	Observed data	What can we infer about the model or process?

For example: In probability, we might assume that a coin is fair and calculate

$$Prob(\text{five heads in five tosses})$$

In statistics, we might observe five heads in five tosses and ask whether this provides evidence that the coin is biased.

Key Takeaways Probability theory provides the mathematical language for describing uncertainty. An event is denoted by A , and its probability is written as

$$Prob(A)$$

A random variable is usually written as a capital letter, such as X , while its realised value is written as a lower-case letter, such as x .

The frequentist interpretation views probability as long-run relative frequency.

The subjective interpretation views probability as a degree of belief.

Probability starts with a model and studies possible outcomes.

Statistics starts with data and makes inference about the model or data-generating process.

Bayesian statistics uses probability to represent uncertainty about unknown quantities.

1.2 Univariate Discrete Distributions

1.2.1 Random Variables and Support

A random variable X assigns numerical values to the outcomes of a random experiment. The support of X , usually denoted by S , is the set of all values that X can take with probability one.

For example, if X is the number of successes in n trials, then the possible values are

$$S = \{0, 1, \dots, n\}.$$

This is because the number of successes cannot be smaller than 0 or greater than n .

A random variable is called discrete if its support is countable. The support may be finite, such as

$$S = \{0, 1, \dots, n\},$$

or countably infinite, such as

$$S = \{0, 1, 2, \dots\},$$

A random variable is called continuous if its possible values form an interval or another uncountable set. For example, the amount of rainfall on a given day may take any non-negative real value, so its support could be

$$S = [0, \infty)$$

1.2.2 Probability Mass Function(PMF)

For a discrete random variable, the probability mass function, or PMF, assigns a probability to each value in the support of X . It is defined as

$$f(x) = \text{Prob}(X = x).$$

A function $f(x)$ is a valid PMF if it satisfies two conditions:

$$f(x) \geq 0 \text{ for all } x \in S$$

and

$$\sum_{x \in S} f(x) = 1.$$

The first condition states that probabilities cannot be negative. The second condition states that the probabilities of all possible outcomes must add up to one. If S' is a subset of the support S , then the probability that X falls in S' is

$$\text{Prob}(X \in S') = \sum_{x \in S'} f(x)$$

For example, if x_1 and x_2 are two possible values of X , then

$$\text{Prob}(X = x_1 \text{ or } X = x_2) = f(x_1) + f(x_2)$$

1.2.3 Cumulative Distribution Function

The cumulative distribution function, or **CDF**, of a random variable X is defined as

$$F(x) = \text{Prob}(X \leq x)$$

For a discrete random variable, the CDF is obtained by summing the PMF over all values less than or equal to x :

$$F(x) = \sum_{x \leq c} f(c)$$

The CDF therefore gives the probability that the random variable takes a value no greater than x .

1.2.4 Mean and Variance

The expected value, or mean, of a discrete random variable is

$$E(X) = \sum_{x \in S} x f(x)$$

The mean describes the centre of the distribution.

The variance measures the spread of the distribution around its mean. It is defined as

$$Var(X) = E[X - E(X)]^2$$

For a discrete random variable, this becomes

$$Var(X) = \sum_{x \in S} [x - E(X)]^2 f(x)$$

A useful equivalent formula is

$$Var(X) = E(X^2) - [E(X)]^2$$

where

$$E(X^2) = \sum_{x \in S} x^2 f(x)$$

The standard deviation is the square root of the variance:

$$SD(x) = \sqrt{Var(X)}$$

It is often easier to interpret than the variance because it is measured on the same scale as X .

1.2.5 Samples, Statistics and Sampling Distributions

In statistical analysis, we often observe a sample from a population. A sample may be written as

$$X_1, \dots, X_n$$

where each X_i is a random variable from the same population distribution.

The sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

It is important to distinguish between the population mean $E(X)$ and the sample mean $\bar{X}$. The population mean is a fixed feature of the distribution, whereas the sample mean depends on the particular sample observed.

A **statistic** is any function of the sample. For example, $\bar{X}$ is a statistic. Since a statistic depends on random data, it is itself a random variable. The probability distribution of a statistic is called its **sampling distribution**.

1.2.6 Parametric Statistical Models

A **parametric model** assumes that the distribution of X belongs to a known family of distributions, but that one or more parameters are unknown.

We often write the PMF as $f(x|\theta)$ which means the probability of observing $X = x$ given the parameter θ . The vertical line in $f(x | \theta)$ should be read as “given”. It is not division. The advantage of a parametric model is that the whole distribution can often be described using only a small number of parameters. Once these parameters are known or estimated, the probabilities, mean, variance and other properties of the distribution can be determined.

1.2.7 Common Discrete Distributions

1.2.7.1 Bernoulli Distribution

A Bernoulli random variable models a single trial with two possible outcomes: success and failure. We usually code success as 1 and failure as 0. If

$$X \in \{0, 1\}$$

and

$$Prob(X = 1 | \theta) = \theta$$

then

$$Prob(X = 0 | \theta) = 1 - \theta$$

We write

$$X \sim Bernoulli(\theta)$$

where

$$0 \leq \theta \leq 1$$

The PMF is

$$f(x | \theta) = \theta^x (1 - \theta)^{1-x}, x = 0, 1$$

This compact formula works because : when $x = 1$:

$$f(1 | \theta) = \theta^1 (1 - \theta)^0 = \theta$$

and when $x = 0$:

$$f(0 | \theta) = \theta^0 (1 - \theta)^1 = 1 - \theta$$

The mean of a Bernoulli random variable is

$$E(X|\theta) = \sum_{x=0}^1 x f(x|\theta) = f(1|\theta) = \theta$$

The variance is $\theta(1 - \theta)$

$$\begin{aligned} \text{Var}(X|\theta) &= \sum_{x=0}^1 [x - E(X|\theta)]^2 f(x|\theta) \\ &= [0 - \theta]^2 \theta^0 (1 - \theta)^{1-0} + [1 - \theta]^2 \theta^{1-0} (1 - \theta)^0 \\ &= \theta^2 - \theta^3 + \theta - 2\theta^2 + \theta^3 \\ &= \theta - \theta^2 \\ &= \theta(1 - \theta) \quad (\text{End Proof}) \end{aligned} \tag{1.2.1}$$

1.2.7.2 Binomial Distribution

The binomial distribution generalises the Bernoulli distribution from one trial to n independent trials. Suppose

$$X_1, \dots, X_n$$

are independent Bernoulli random variables with success probability θ . That is,

$$X_i \sim \text{Bernoulli}(\theta), i = 1, \dots, n.$$

Let

$$X = \sum_{i=1}^n X_i$$

Then X is the total number of successes in n independent trials, and

$$X \sim \text{Binomial}(n, \theta)$$

The support is

$$S = \{0, 1, \dots, n\}$$

The PMF is

$$f(x | \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, x = 0, 1, \dots, n.$$

Here,

$$\binom{n}{x}$$

is the binomial coefficient, which counts the number of ways to choose x successes from n trials.

The mean is

$$E(X | \theta) = n\theta$$

The variance is

$$\text{Var}(X | \theta) = n\theta(1 - \theta)$$

This follows from the fact that X is the sum of n independent Bernoulli random variables. Since each Bernoulli random variable has mean θ ,

$$E(X | \theta) = E\left(\sum_{i=1}^n X_i | \theta\right) = \sum_{i=1}^n E(X_i | \theta) = n\theta$$

Since the X_i 's are independent,

$$\text{Var}(X | \theta) = \sum_{i=1}^n \text{Var}(X_i | \theta) = \sum_{i=1}^n \theta(1 - \theta) = n\theta(1 - \theta)$$

1.2.7.3 Poisson Distribution

The Poisson distribution is commonly used to model count data, especially the number of events occurring in a fixed interval of time or space.

Examples include:

- the number of calls received by a call centre in one hour;
- the number of accidents at a junction in one month;
- the number of emails received in a day;
- the number of events occurring in a fixed observation period.

If

$$X \sim \text{Poisson}(\theta)$$

then the support is

$$S = \{0, 1, 2, \dots\}$$

where

$$\theta \geq 0,$$

The PMF is

$$f(x | \theta) = \frac{e^{-\theta} \theta^x}{x!}, x = 0, 1, 2, \dots$$

The mean and variance of a Poisson random variable are both equal to θ :

$$E(X | \theta) = \theta$$

and

$$Var(X | \theta) = \theta$$

This equality between the mean and variance is an important property of the Poisson distribution.

Distribution	Support	PMF	Mean	Variance
Bernoulli(θ)	$\{0, 1\}$	$\theta^x (1 - \theta)^{1-x}$	θ	$\theta(1 - \theta)$
Binomial(n, θ)	$\{0, 1, \dots, n\}$	$\binom{n}{x} \theta^x (1 - \theta)^{n-x}$	$n\theta$	$n\theta(1 - \theta)$
Poisson(θ)	$\{0, 1, 2, \dots\}$	$\frac{e^{-\theta} \theta^x}{x!}$	θ	θ

Table 1.1: Summary of Common Discrete Distributions

1.2.7.4 Key Takeaways

A discrete random variable has a countable support.

The PMF gives the probability of each possible value:

$$f(x) = Prob(X = x)$$

A valid PMF must be non-negative and sum to one.

The CDF gives cumulative probability:

$$F(x) = Prob(X \leq x).$$

The mean and variance of a discrete random variable can be calculated from its PMF:

$$E(X) = \sum_{x \in S} x f(x) \quad Var(X) = \sum_{x \in S} [x - E(X)]^2 f(x)$$

The Bernoulli distribution models one binary trial.

The binomial distribution models the number of successes in n independent Bernoulli trials.

The Poisson distribution models counts of events in a fixed interval.

In parametric statistics, we write

$$f(x | \theta)$$

to show that the distribution depends on an unknown parameter θ . The aim of statistical inference is often to estimate this parameter from data.

1.2.8 Continuous Distributions

For continuous random variables, the probability mass function does not apply. This is because a continuous random variable can take uncountably many values.

For example, suppose X represents daily rainfall in inches. Then X can take any non-negative real value, so its support may be

$$S = [0, \infty)$$

For a continuous random variable, the probability of observing any exact value is zero:

$$Prob(X = x) = 0$$

This does not mean that the value is impossible. Rather, it means that probability for continuous variables must be assigned to intervals, not individual points.

For example, instead of asking for

$$Prob(X = \pi/2),$$

we ask for the probability that X lies in a small interval around $\pi/2$, such as

$$Prob(\pi/2 - \epsilon < X < \pi/2 + \epsilon)$$

1.2.8.1 Cumulative Distribution Function

For both discrete and continuous random variables, the cumulative distribution function is defined as

$$F(x) = Prob(X \leq x).$$

For continuous random variables, the CDF is especially important because probabilities are usually obtained from differences in the CDF. For example,

$$Prob(a \leq X \leq b) = F(b) - F(a).$$

This gives the probability that X falls inside the interval $[a, b]$.

1.2.8.2 Probability Density Function(PDF)

For a continuous random variable, probabilities are described using a probability density function, or **PDF**, denoted by $f(x)$.

The PDF is related to the CDF by differentiation:

$$f(x) = F'(x).$$

Equivalently, the CDF can be written as:

$$F(x) = \int_{-\infty}^x f(t)dt.$$

The probability that X lies between a and b is the area under the density curve between a and b :

$$Prob(a \leq X \leq b) = F(b) - F(a) = \int_a^b f(x)dx$$

For a very small interval around x ,

$$Prob(x - \epsilon < X < x + \epsilon) \approx 2\epsilon f(x).$$

This approximation explains why the PDF is useful: it describes how probability is locally concentrated around a value.

1.2.8.3 Validity Conditions for a PDF

A function $f(x)$ is a valid PDF if it satisfies two conditions:

$$f(x) \geq 0 \text{ for all } x,$$

and

$$\int_{-\infty}^{\infty} f(x)dx = 1,$$

The first condition means that density cannot be negative. The second condition means that the total area under the density curve must equal one.

A key point is that $f(x)$ is not itself a probability. It is a density. Therefore, it is possible for $f(x)$ to be greater than one for some values of x , as long as the total area under the curve is one.

1.2.8.4 Mean and Variance of a Continuous Random Variable

The formulas for the mean and variance of a continuous random variable are similar to those for a discrete random variable. The main difference is that sums are replaced by integrals.

The mean is

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

The variance is

$$Var(X) = E\{[X - E(X)]^2\}$$

For a continuous random variable, this becomes

$$Var(X) = \int_{-\infty}^{\infty} [x - E(X)]^2 f(x)dx.$$

A useful equivalent formula is

$$Var(X) = E(X^2) - [E(X)]^2,$$

where

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx.$$

1.2.8.5 Quantiles

For $\tau \in [0, 1]$, the τ -quantile of a continuous random variable is the value $Q(\tau)$ such that

$$Prob(X \leq Q(\tau)) = \tau.$$

Equivalently,

$$F(Q(\tau)) = \tau$$

Therefore,

$$Q(\tau) = F^{-1}(\tau).$$

The median is the 0.5-quantile:

$$Q(0.5)$$

A central equal-tailed interval with probability $1 - \alpha$ is

$$[Q(\alpha/2), Q(1 - \alpha/2)].$$

This interval satisfies

$$Prob\{Q(\alpha/2) \leq X \leq Q(1 - \alpha/2)\} = 1 - \alpha.$$

1.2.9 Common Continuous Distributions

1.2.9.1 Gaussian Distribution

The Gaussian distribution, also called the normal distribution, is one of the most important continuous distributions. If

$$X \sim N(\mu, \sigma^2)$$

then X has support

$$S = (-\infty, \infty).$$

The PDF is

$$f(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(x - \mu)^2\right].$$

The mean and variance are

$$E(X | \mu, \sigma^2) = \mu,$$

and

$$Var(X | \mu, \sigma^2) = \sigma^2.$$

The Gaussian distribution is important because of the central limit theorem. Under fairly general conditions, the distribution of the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is approximately normal for large n , even if the original data are not normally distributed.

1.2.9.2 Gamma Distribution

The Gamma distribution is a flexible distribution for positive continuous random variables. If

$$X \sim \text{Gamma}(a, b),$$

using the shape-rate parameterisation, then the support is

$$S = [0, \infty).$$

The PDF is

$$f(x | \theta) = \begin{cases} \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx), & x \geq 0, \\ 0, & x < 0, \end{cases}$$

where

$$a > 0, b > 0.$$

Here, a is the shape parameter and b is the rate parameter.

The mean and variance are

$$E(X | a, b) = \frac{a}{b},$$

and

$$Var(X | a, b) = \frac{a}{b^2}.$$

1.2.9.3 Beta Distribution

The Beta distribution is a flexible distribution on the interval $[0, 1]$. It is especially useful for modelling probabilities and proportions. If

$$X \sim \text{Beta}(a, b),$$

then the support is

$$S = [0, 1].$$

The PDF is

$$f(x | \theta) = \begin{cases} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, & x \in [0, 1], \\ 0, & x < 0 \text{ or } x > 1. \end{cases}$$

where

$$a > 0, b > 0.$$

The Beta distribution can take many shapes depending on a and b . It can be left-skewed, right-skewed, symmetric, U-shaped or approximately uniform. When

$$a = b = 1,$$

the Beta distribution becomes the uniform distribution on $[0,1]$:

$$f(x) = 1, 0 \leq x \leq 1.$$

The mean and variance are

$$E(X | a, b) = \frac{a}{a + b},$$

and

$$Var(X | a, b) = \frac{ab}{(a + b)^2(a + b + 1)},$$

Distribution	Support	PMF	Mean	Variance
Gaussian $N(\mu, \sigma^2)$	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp[-\frac{(x-\mu)^2}{2\sigma^2}]$	μ	σ^2
Gamma(a, b)	$[0, \infty)$	$\frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$	$\frac{a}{b}$	$\frac{a}{b^2}$
Beta(a, b)	$[0, 1]$	$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}$	$\frac{a}{a+b}$	$\frac{ab}{(a+b)^2(a+b+1)}$

Table 1.2: Summary of Common Continuous Distributions

1.2.9.4 Key Takeaways

For continuous random variables,

$$\Pr(X = x) = 0.$$

Therefore, probabilities are assigned to intervals rather than exact points.

The cumulative distribution function is

$$F(x) = \Pr(X \leq x).$$

The probability density function is the derivative of the CDF:

$$f(x) = F'(x).$$

Interval probabilities are calculated by integration:

$$\Pr(a \leq X \leq b) = \int_a^b f(x) dx.$$

A valid PDF must satisfy

$$f(x) \geq 0$$

and

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

For continuous random variables, sums are replaced by integrals:

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx,$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} [x - E(X)]^2 f(x) dx.$$

The Gaussian distribution is used widely because of the central limit theorem.

The Gamma distribution is useful for positive continuous variables.

The Beta distribution is useful for variables restricted to $[0, 1]$, especially probabilities and proportions.

1.3 Multivariate Distribution

Most statistical analyses involve more than one variable. In many cases, the aim is not only to describe each variable separately, but also to understand the relationship between variables.

For example, in a medical study, one variable may indicate whether a patient has a primary health condition, while another variable may indicate whether the patient experiences a side effect. To study the relationship between these variables, we need a multivariate distribution.

Let

$$X_1, \dots, X_p$$

be p random variables. The random vector is written as

$$\mathbf{X} = (X_1, \dots, X_p)$$

If S_j is the support of X_j , then

$$X_j \in S_j, j = 1, \dots, p.$$

The joint distribution of $\mathbf{X}$ describes how the variables behave together.

1.3.1 Joint Probability Mass Function

If all variables are discrete, their joint probability mass function, or joint PMF, is defined as

$$\text{Prob}(X_1 = x_1, \dots, X_p = x_p) = f(x_1, \dots, x_p)$$

This gives the probability that all variables take specific values at the same time. For $f(x_1, \dots, x_p)$ to be a valid joint PMF, it must satisfy two conditions:

$$f(x_1, \dots, x_p) \geq 0$$

for all possible values, and

$$\sum_{x_1 \in S_1} \dots \sum_{x_p \in S_p} f(x_1, \dots, x_p) = 1$$

The first condition states that probabilities cannot be negative. The second condition states that the probabilities of all possible combinations of outcomes must add up to one.

If $\mathbf{A}$ is a set of possible values of the random vector $\mathbf{X}$, then

$$\text{Prob}(\mathbf{X} \in \mathbf{A}) = \sum_{(x_1, \dots, x_p) \in \mathbf{A}} f(x_1, \dots, x_p),$$

So, for discrete multivariate random variables, probabilities are calculated by summing the joint PMF over the relevant combinations of values.

1.3.1.1 Joint Probability Density Function

If the variables are continuous, probabilities are described using a joint probability density function, or joint PDF.

For two continuous random variables X_1 and X_2 , the joint PDF is written as

$$f(x_1, x_2).$$

The probability that X_1 lies between a_1 and b_1 , and X_2 lies between a_2 and b_2 , is:

$$Prob(a_1 < X_1 < b_1, a_2 < X_2 < b_2) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x_1, x_2) dx_2 dx_1,$$

This probability is the volume under the joint density surface over the rectangle defined by the intervals

$$(a_1, b_1) \text{ and } (a_2, b_2).$$

For p continuous random variables, the probability that the random vector X falls in a region A is

$$Prob(X \in A) = \int_A f(x_1, \dots, x_p) dx_1 \dots dx_p.$$

Thus, in the multivariate continuous case, probabilities are computed by integrating the joint PDF over a region.

1.3.1.2 Validity Conditions for a Joint PDF

A function $f(x_1, \dots, x_p)$ is a valid joint PDF if

$$f(x_1, \dots, x_p) \geq 0$$

for all values, and

$$\int_{\mathbb{R}^p} f(x_1, \dots, x_p) dx_1 \dots dx_p = 1$$

If the support is smaller than $\mathbb{R}^p$, the integral is taken over the support of the random vector.

As in the univariate continuous case, the value of a joint PDF is not itself a probability. Probabilities are obtained by integrating the density over a region.

1.3.1.3 Key Takeaways

A multivariate distribution describes several random variables jointly.

A random vector is written as

$$X = (X_1, \dots, X_p).$$

For discrete random variables, the joint PMF is

$$f(x_1, \dots, x_p) = \Pr(X_1 = x_1, \dots, X_p = x_p).$$

A valid joint PMF must be non-negative and sum to one:

$$\sum_{x_1 \in S_1} \dots \sum_{x_p \in S_p} f(x_1, \dots, x_p) = 1.$$

For continuous random variables, probabilities are calculated using a joint PDF.

For two variables,

$$\Pr(a_1 < X_1 < b_1, a_2 < X_2 < b_2) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x_1, x_2) dx_2 dx_1.$$

For p variables,

$$\Pr(\mathbf{X} \in A) = \int_A f(x_1, \dots, x_p) dx_1 \cdots dx_p.$$

The main idea is:

univariate probability: area under a curve

bivariate probability: volume under a surface

multivariate probability: integral over a region

1.4 Marginal and Conditional Distributions

A multivariate distribution describes several random variables jointly. From a joint distribution, we can derive **marginal distributions** and **conditional distributions**.

Marginal and conditional distributions are especially important in Bayesian statistics because Bayesian analysis often works with joint models and then studies individual variables, parameters or conditional relationships.

To introduce the ideas, consider two discrete random variables,

$$X_1 \text{ and } X_2$$

with joint PMF

$$f(x_1, x_2) = \text{Prob}(X_1 = x_1, X_2 = x_2)$$

1.4.1 Marginal Distributions

For X_1 , the marginal PMF is

$$f_1(x_1) = \sum_{x_2} f(x_1, x_2).$$

For X_2 , the marginal PMF is

$$f_2(x_2) = \sum_{x_1} f(x_1, x_2).$$

The word “marginal” comes from two-way probability tables: the marginal probabilities appear in the margins, as row totals and column totals.

For example, if the joint PMF is displayed in a table, then the row totals give the marginal distribution of X_1 , while the column totals give the marginal distribution of X_2 .

1.4.2 Marginal Mean and Variance

Once we have the marginal distribution of X_j , its mean is

$$\mu_j = E(X_j) = \sum_{x_j \in S_j} x_j f_j(x_j).$$

Equivalently, using the joint PMF for two variables,

$$\mu_j = \sum_{x_1} \sum_{x_2} x_j f(x_1, x_2).$$

The marginal variance is

$$\sigma_j^2 = \text{Var}(X_j) = \sum_{x_j} (x_j - \mu_j)^2 f_j(x_j).$$

Equivalently,

$$\sigma_j^2 = \sum_{x_1} \sum_{x_2} (x_j - \mu_j)^2 f(x_1, x_2).$$

The marginal mean and variance describe the centre and spread of each individual variable. However, they do not describe the relationship between the variables.

1.4.3 Covariance and Correlation

To summarise the relationship between two random variables, we often use covariance.

The covariance between X_1 and X_2 is

$$\sigma_{12} = \text{Cov}(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)].$$

For discrete variables,

$$\sigma_{12} = \sum_{x_1} \sum_{x_2} (x_1 - \mu_1)(x_2 - \mu_2) f(x_1, x_2).$$

An equivalent and often useful formula is

$$\text{Cov}(X_1, X_2) = E(X_1 X_2) - E(X_1)E(X_2).$$

Covariance depends on the measurement scales of X_1 and X_2 , so it can be difficult to interpret directly. A scale-free measure of linear association is the correlation:

$$\rho_{12} = \text{Cor}(X_1, X_2) = \frac{\sigma_{12}}{\sigma_1 \sigma_2}.$$

The correlation always lies between -1 and 1 .

1.4.4 Mean Vector and Covariance Matrix

For a random vector

$$\mathbf{X} = (X_1, X_2)^T,$$

the mean vector is

$$E(\mathbf{X}) = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}.$$

The covariance matrix is

$$\text{Cov}(\mathbf{X}) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}.$$

The diagonal elements are variances, and the off-diagonal elements are covariances.

For p variables,

$$\mathbf{X} = (X_1, \dots, X_p)^T,$$

the mean vector is

$$E(\mathbf{X}) = (\mu_1, \dots, \mu_p)^T,$$

and the covariance matrix is a symmetric $p \times p$ matrix. Its diagonal elements are variances and its off-diagonal elements are covariances.

1.4.5 Conditional Distributions

A conditional distribution describes the distribution of one variable after another variable has been fixed or observed.

For example, the conditional PMF of X_1 given $X_2 = x_2$ is

$$f_{1|2}(x_1 | X_2 = x_2) = \frac{f(x_1, x_2)}{f_2(x_2)}.$$

Similarly,

$$f_{2|1}(x_2 | X_1 = x_1) = \frac{f(x_1, x_2)}{f_1(x_1)}.$$

The denominator rescales the probabilities so that the conditional probabilities sum to one. This is the discrete version of the general conditional probability rule:

$$\Pr(A | B) = \frac{\Pr(A \cap B)}{\Pr(B)}.$$

1.4.6 Key Takeaways

The marginal distribution of one variable is obtained by summing the joint distribution over the other variable.

$$f_1(x_1) = \sum_{x_2} f(x_1, x_2), \quad f_2(x_2) = \sum_{x_1} f(x_1, x_2).$$

Marginal means and variances describe individual variables, but not the relationship between variables. Covariance and correlation describe association between variables.

$$\text{Cov}(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)].$$

$$\text{Cor}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sigma_1 \sigma_2}.$$

Conditional distributions describe the distribution of one variable after another variable has been fixed or observed.

$$f_{1|2}(x_1 | X_2 = x_2) = \frac{f(x_1, x_2)}{f_2(x_2)}.$$

This section is important because Bayesian inference is built around joint, marginal and conditional distributions.

1.5 Bayes' Rule

Bayes' rule is a fundamental result in probability theory and the foundation of Bayesian statistics. It provides a way to reverse conditional probabilities.

Suppose X_1 and X_2 are two random variables with joint distribution

$$f(x_1, x_2).$$

From the definition of conditional distributions,

$$f(x_1 | x_2) = \frac{f(x_1, x_2)}{f_2(x_2)}$$

and

$$f(x_2 | x_1) = \frac{f(x_1, x_2)}{f_1(x_1)}.$$

Since the joint distribution can be written as

$$f(x_1, x_2) = f(x_1 | x_2) f_2(x_2),$$

we obtain Bayes' rule:

$$f(x_2 | x_1) = \frac{f(x_1 | x_2)f(x_2)}{f(x_1)}.$$

In Bayesian statistics, this is usually interpreted as

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{marginal likelihood}}$$

Equivalently,

$$f(\theta | y) = \frac{f(y | \theta)f(\theta)}{f(y)}.$$

Here:

$f(\theta | y)$ is the **posterior distribution** of the unknown parameter after observing data;

$f(y | \theta)$ is the **likelihood**, the distribution of the data given the parameter;

$f(\theta)$ is the **prior distribution**, representing uncertainty about the parameter before observing the data;

$f(y)$ is the **marginal likelihood** or **evidence**, which ensures that the posterior distribution integrates or sums to one.

1.5.1 Discrete Example of Bayes' Rule

Consider a medical testing example. Let $Y \in \{0, 1\}$ be the test result, where $Y = 1$ means the test is positive, and $Y = 0$ means the test is negative. Let $\theta \in \{0, 1\}$ be the true disease status, where $\theta = 1$ means the patient truly has the disease, and $\theta = 0$ means the patient does not have the disease.

The test is not perfect. Define:

$$p = \Pr(Y = 1 | \theta = 0),$$

the false positive rate, and

$$q = \Pr(Y = 0 | \theta = 1),$$

the false negative rate.

Therefore,

$$\Pr(Y = 1 | \theta = 1) = 1 - q,$$

and

$$\Pr(Y = 0 | \theta = 0) = 1 - p.$$

The likelihood can be summarised as follows:

	$Y = 1$	$Y = 0$
$\theta = 1$	$1 - q$	q
$\theta = 0$	p	$1 - p$

Let $\pi = \Pr(\theta = 1)$ be the prior probability that the patient has the disease before observing the test result. Then

$$\Pr(\theta = 0) = 1 - \pi.$$

Suppose the test result is positive, so $Y = 1$. Bayes' rule gives

$$\Pr(\theta = 1 | Y = 1) = \frac{\Pr(Y = 1 | \theta = 1) \Pr(\theta = 1)}{\Pr(Y = 1)}.$$

The denominator is the marginal probability of a positive test:

$$\Pr(Y = 1) = \Pr(Y = 1 | \theta = 1) \Pr(\theta = 1) + \Pr(Y = 1 | \theta = 0) \Pr(\theta = 0).$$

Substituting the terms gives

$$\Pr(Y = 1) = (1 - q)\pi + p(1 - \pi).$$

Therefore, the posterior probability of having the disease after a positive test is

$$\Pr(\theta = 1 \mid Y = 1) = \frac{(1 - q)\pi}{(1 - q)\pi + p(1 - \pi)}.$$

Similarly, after a negative test,

$$\Pr(\theta = 1 \mid Y = 0) = \frac{q\pi}{q\pi + (1 - p)(1 - \pi)}.$$

Interpretation:

This example separates two types of uncertainty.

The test result Y is random before the test is taken. If the test were repeated under similar conditions, different outcomes might occur. This is aleatoric uncertainty, meaning randomness arising from the data-generating process.

The disease status θ may be fixed in reality, but unknown to us. Our uncertainty about θ is **epistemic uncertainty**, meaning uncertainty due to lack of knowledge.

Bayesian statistics represents uncertainty about unknown quantities using probability distributions. In this example, the prior distribution represents uncertainty about θ before observing the test result, and the posterior distribution represents uncertainty about θ after observing the test result.

Extreme Cases:

Several special cases help interpret the posterior. If $\pi = 1$, then the prior probability of disease is already one, and the posterior remains one regardless of the test result. If $\pi = 0$, then the prior probability of disease is zero, and the posterior remains zero regardless of the test result. If the test is perfect, so that

$$p = 0 \quad \text{and} \quad q = 0,$$

then a positive test implies

$$\Pr(\theta = 1 \mid Y = 1) = 1,$$

and a negative test implies

$$\Pr(\theta = 1 \mid Y = 0) = 0.$$

If the test is completely uninformative, for example

$$p = q = \frac{1}{2},$$

then the posterior equals the prior:

$$\Pr(\theta = 1 \mid Y) = \pi.$$

In this case, the data provide no useful information about θ .

1.5.2 Continuous Example of Bayes' Rule

Now consider a continuous parameter. Let $\theta \in [0, 1]$ be the proportion of people in a county who have health insurance. Suppose the prior distribution of θ is

$$\theta \sim \text{Beta}(a, b).$$

The prior PDF is

$$f(\theta) = \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1}, \quad 0 \leq \theta \leq 1.$$

Now suppose we observe a sample of size n , and let Y be the number of people in the sample who have health insurance. Conditional on θ ,

$$Y \mid \theta \sim \text{Binomial}(n, \theta).$$

The likelihood is

$$f(y | \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}.$$

Bayes' rule gives

$$f(\theta | y) = \frac{f(y | \theta) f(\theta)}{f(y)}.$$

To derive the posterior, multiply the likelihood and the prior:

$$f(y | \theta) f(\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1}.$$

Ignoring constants that do not depend on θ , we have

$$f(\theta | y) \propto \theta^{y+a-1} (1 - \theta)^{n-y+b-1}.$$

This has the same form as a Beta distribution. Therefore,

$$\theta | Y = y \sim \text{Beta}(y + a, n - y + b).$$

This is an example of **conjugacy**. The Beta prior and the Binomial likelihood produce a Beta posterior. The posterior parameters are

$$a_{\text{post}} = a + y,$$

and

$$b_{\text{post}} = b + n - y.$$

Thus, the observed number of successes y is added to the first Beta parameter, and the observed number of failures $n - y$ is added to the second Beta parameter. A useful posterior mean is

$$\mathbb{E}(\theta | Y = y) = \frac{a + y}{a + b + n}.$$

This shows directly how Bayesian updating combines prior information and observed data.

1.5.3 The Role of the Marginal Likelihood

The marginal likelihood is

$$f(y) = \int f(y | \theta) f(\theta) d\theta.$$

It acts as a normalising constant, ensuring that the posterior density integrates to one.

In many posterior derivations, we do not need to compute this constant directly. Instead, we use proportionality:

$$f(\theta | y) \propto f(y | \theta) f(\theta).$$

Then we identify the resulting expression as a known distribution.

For the Beta-Binomial model,

$$f(\theta | y) \propto \theta^{y+a-1} (1 - \theta)^{n-y+b-1},$$

so we recognise the posterior as

$$\text{Beta}(y + a, n - y + b).$$

1.5.4 Key Takeaways

Bayes' rule reverses conditioning:

$$f(\theta | y) = \frac{f(y | \theta) f(\theta)}{f(y)}.$$

In Bayesian statistics,

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

The prior represents uncertainty before observing data.

The likelihood describes how probable the observed data are under different parameter values.

The posterior represents updated uncertainty after observing data.

The marginal likelihood normalises the posterior.

In the discrete medical testing example,

$$\Pr(\theta = 1 \mid Y = 1) = \frac{(1 - q)\pi}{(1 - q)\pi + p(1 - \pi)}.$$

In the continuous Beta-Binomial example,

$$\theta \sim \text{Beta}(a, b), \quad Y \mid \theta \sim \text{Binomial}(n, \theta),$$

implies

$$\theta \mid Y = y \sim \text{Beta}(y + a, n - y + b).$$

This section is essential because almost all Bayesian modelling follows the same structure:

$\text{prior} + \text{data through likelihood} \Rightarrow \text{posterior}$

1.6 Introduction to Bayesian inference